MEMORANDUM

TO: Professor Brodnax
FROM: Guo Zhan
DATE: October 30, 2026
RE: Memo 2: Concept Showcase Plan

Concept

In my analysis, artemether-lumefantrine, the largest malaria medicine at USD 0.9 billion, sits among HIV generics in the product network, not with other malaria medicines. That placement raises a question for a buyer: if disease area does not show who could make a medicine, what does?

My showcase will highlight the results of my Insight Report, and the answer it offers is shared suppliers. It will present two findings. First, medicines cluster by the firms that supply them. Louvain community detection (Blondel et al., 2008) found seven communities with a modularity of 0.31, above the 0.22 average from 200 shuffled networks, and these communities only partly match disease areas. Second, this structure helps predict who enters next. Density (Hidalgo et al., 2007) measures how close a medicine is to what a firm already supplies. Three-year entry rose from 0.15% at zero density to 8.1% above 0.5. On later test years, adding density raised the share of actual entries found in the top 5% of predictions from 35% to 47%.

These findings matter because 39 medicines bought in 2021-2025 relied on one or two manufacturers. For 31 of them, my model can rank likely new suppliers. For a medicine such as pyronaridine-artesunate, density is zero for every candidate, so a buyer cannot expect the network to point to an entrant. The audience is a market-shaping lead deciding where to spend limited effort.

Medium

The final product will be an interactive website, The Second Supplier Map, built with D3.js from my output tables and submitted as a web link. A 155-row candidate table is hard to use in a meeting, while a map lets a reader start from one medicine and follow its connections. The site will have three parts:

1. A guided scroll. Medicines first appear grouped by disease area, then move into their network communities as the reader scrolls, so misfits such as artemether-lumefantrine become visible.

2. A guessing game. The reader picks a medicine with few suppliers and guesses which firm could enter next, then sees the model's ranking. For bedaquiline, supplied by Janssen and Lupin, Macleods ranks first. This makes the idea of density concrete before any formula appears.

3. A supplier finder. The reader can select any of the 31 medicines to see current suppliers, the top five candidates, their density, and whether a text match suggests a WHO prequalification listing. Where density is zero, the site will say that the network offers no signal.

The site will state its limits next to the results. Predictions rest on only 34 test entries, so I will present candidates as a ranking to investigate, not a forecast. PQR also covers only Global Fund-financed purchases, and the WHO flag needs checking by hand.

I chose this medium because it matches the question a buyer actually asks: not how the network works, but who to call about this medicine.

References

Blondel, V. D., Guillaume, J.-L., Lambiotte, R., & Lefebvre, E. (2008). Fast unfolding of communities in large networks. *Journal of Statistical Mechanics: Theory and Experiment, 2008*(10), P10008. https://doi.org/10.1088/1742-5468/2008/10/P10008

Hidalgo, C. A., Klinger, B., Barabási, A.-L., & Hausmann, R. (2007). The product space conditions the development of nations. *Science, 317*(5837), 482–487. https://doi.org/10.1126/science.1144581